Lecture Notes on Optimization and Optimal Control

Module 2: Classical Optimality Conditions

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This lecture develops the analytical foundations of constrained optimization, with emphasis on first- and second-order optimality conditions for nonlinear programs involving equality and inequality constraints. Beginning with the geometric notion of feasibility and local optimality, we derive the Fritz–John necessary conditions using theorems of the alternative and clarify the role of constraint qualifications in recovering the Karush–Kuhn–Tucker (KKT) framework. Particular attention is paid to the Mangasarian–Fromovitz constraint qualification and its function in excluding degenerate multiplier configurations.

The lecture then advances to second-order analysis, introducing critical directions and second-order tangent behavior as the appropriate tools for characterizing curvature of the objective along feasible directions. We present both second-order necessary and sufficient conditions in Fritz–John form, highlighting their inherently directionwise nature and the subtle interaction between active constraints and curvature. The distinction between local, strict, and isolated local optima is examined in detail, with illustrative examples demonstrating how first- and second-order conditions may hold even when a point fails to be locally optimal.

Throughout, the lecture emphasizes geometric intuition, structural limitations of derivative-based conditions, and the precise circumstances under which second-order conditions guarantee isolation of a local optimum. These results provide the theoretical backbone for modern nonlinear programming methods and motivate algorithmic approaches such as sequential quadratic programming.

1 Constrained Optimization and Optimality Concepts

We consider a general constrained optimization problem in which the feasible set is specified algebraically by a collection of equality and inequality constraints:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}. \end{aligned} \tag{1.1}$$

Here, $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function to be minimized, $h_i : \mathbb{R}^n \to \mathbb{R}$ represent equality constraints indexed by the finite set $\mathcal{P}$, and $g_j : \mathbb{R}^n \to \mathbb{R}$ represent inequality constraints indexed by the finite set $\mathcal{Q}$.

The *feasible set* associated with (1.1) is defined as

$$\mathcal{F} = \{\, x \in \mathbb{R}^n \mid h_i(x) = 0,\ i \in \mathcal{P},\ g_j(x) \leq 0,\ j \in \mathcal{Q} \,\}.$$

Any point $x \in \mathcal{F}$ is called a *feasible point*. Points not belonging to $\mathcal{F}$ violate at least one constraint and are therefore infeasible. Problems of the form (1.1) are commonly referred to as *nonlinear programs* when at least one of the functions f, h_i, or g_j is nonlinear.

1.1 Optimality Notions

We now introduce standard notions of optimality for constrained optimization problems.

Global minimum. A feasible point $x^\star \in \mathcal{F}$ is called a *global minimum* (or *globally optimal solution*) of (1.1) if

$$f(x^\star) \leq f(x), \quad \forall x \in \mathcal{F}.$$

The value $f(x^\star)$ is referred to as the *global optimal value*.

Local minimum. A feasible point $x^\star \in \mathcal{F}$ is called a *local minimum* if there exists a neighborhood $N(x^\star)$ of $x^\star$ such that

$$f(x^\star) \leq f(x), \quad \forall x \in \mathcal{F} \cap N(x^\star).$$

A feasible point $x^\star \in \mathcal{F}$ is called an *isolated local minimum* if there exists a neighborhood

$N(x^\star)$ of $x^\star$ such that

$$f(x^\star) < f(x), \qquad \forall x \in \mathcal{F} \cap N(x^\star),\ x \neq x^\star.$$

Every global minimum is also a local minimum, but the converse need not hold. In constrained optimization, local optimality depends on the geometry of the feasible set near the candidate point, rather than on the behavior of the objective function over the entire space $\mathbb{R}^n$.

1.2 Regularity Conditions

1.2.1 A strict hierarchy of regularity

The analysis of constrained optimization problems relies on different levels of smoothness of the objective and constraint functions. These notions form a strict hierarchy of regularity:

$$f \in C^1(U) \implies f \text{ is differentiable on } U \implies f \text{ is continuous on } U,$$

where the converses do not generally hold. Each level of regularity enables progressively stronger analytical tools.

Continuity is the weakest regularity notion and ensures that small perturbations of the argument produce small changes in the function value.

Definition 1.1 (Continuity)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *continuous* at a point $x \in \mathbb{R}^n$ if

$$\lim_{y \to x} f(y) = f(x),$$

that is, for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|y - x\| < \delta \quad \implies \quad |f(y) - f(x)| < \varepsilon.$$

The function f is said to be *continuous on a set* $U \subset \mathbb{R}^n$ if it is continuous at every point in U.

Differentiability strengthens continuity by requiring the existence of a linear approximation that locally captures the behavior of the function.

Definition 1.2 (Differentiability)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *differentiable* at a point

$x \in \mathbb{R}^n$ if there exists a vector $\nabla f(x) \in \mathbb{R}^n$ such that

$$\lim_{\|h\| \to 0} \frac{f(x+h) - f(x) - \nabla f(x)^\top h}{\|h\|} = 0.$$

The vector $\nabla f(x)$ is called the *gradient* of f at x.

Differentiability guarantees a first-order Taylor expansion at the point of interest, but does not impose any regularity on how the gradient behaves in a neighborhood of that point.

Continuous differentiability requires that the gradient itself varies smoothly.

Definition 1.3 (Continuous differentiability)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *continuously differentiable* on an open set $U \subset \mathbb{R}^n$ if it is differentiable at every point $x \in U$ and the gradient mapping

$$x \mapsto \nabla f(x)$$

is continuous on U. In this case, we write $f \in C^1(U)$.

Continuous differentiability ensures the stability of first-order approximations under small perturbations of the expansion point. In particular, it justifies the use of Taylor expansions with controlled remainder terms and is the standard smoothness assumption in first-order optimality theory.

1.2.2 Regularity of constrained optimization problems.

We now translate these smoothness notions into regularity conditions for constrained optimization problems.

Definition 1.4 (First-order regularity)**.** The constrained optimization problem (1.1) is said to be *first-order regular* at a feasible point $x^\star \in \mathcal{F}$ if the objective function f and all constraint functions h_i, $i \in \mathcal{P}$, and g_j, $j \in \mathcal{Q}$, are continuously differentiable in an open neighborhood of $x^\star$.

First-order regularity ensures that gradients of the objective and constraint functions are well defined and vary continuously near $x^\star$, allowing the formulation of feasible directions, constraint qualifications, and first-order optimality conditions such as the Fritz–John and Karush–Kuhn–Tucker conditions.

Definition 1.5 (Second-order regularity)**.** The constrained optimization problem (1.1) is said to be *second-order regular* at a feasible point $x^\star \in \mathcal{F}$ if the objective function f and all constraint functions h_i, $i \in \mathcal{P}$, and g_j, $j \in \mathcal{Q}$, are twice continuously differentiable in an open neighborhood of $x^\star$.

Second-order regularity is required for the formulation of second-order necessary and sufficient optimality conditions, which distinguish local minima from local maxima and saddle points.

1.3 Active and Inactive Constraints

To analyze local optimality conditions, it is essential to distinguish between constraints that are binding at a feasible point and those that are not.

Equality constraints as hard constraints. For every equality constraint

$$h_i(x) = 0, \quad i \in \mathcal{P},$$

feasibility requires that $h_i(x^\star) = 0$ for any feasible point $x^\star$. Hence, *all equality constraints are active at every feasible point*. Equality constraints must be satisfied exactly and impose hard geometric restrictions on the feasible set. In particular, feasible points lie on a lower-dimensional manifold in $\mathbb{R}^n$, and admissible local perturbations must remain tangent to this manifold.

Inequality constraints and the active set. Inequality constraints may or may not influence the local structure of the feasible set.

Definition 1.6 (Active inequality constraints)**.** Let $x^\star \in \mathcal{F}$. An inequality constraint $g_j(x) \leq 0$ is said to be *active* at $x^\star$ if $g_j(x^\star) = 0$, and *inactive* if $g_j(x^\star) < 0$. The set of active inequality constraints at $x^\star$ is defined as

$$\mathcal{Q}(x^\star) = \{\, j \in \mathcal{Q} \mid g_j(x^\star) = 0 \,\}.$$

If an inequality constraint is inactive at $x^\star$, then by continuity there exists a neighborhood of $x^\star$ in which the constraint remains strictly satisfied. Such constraints therefore impose no restriction on sufficiently small feasible perturbations and play no role in local optimality analysis.

Theorem 1.7 (Characterization of Feasible Directions)**.** *Consider the constrained optimization problem* (1.1)*, and let $x^\star \in \mathcal{F}$ be a feasible point. If all constraint functions are differentiable at $x^\star$, then a*

vector $d \in \mathbb{R}^n$ is a feasible direction at $x^\star$ only if

$$\begin{aligned} \nabla h_i(x^\star)^\top d &= 0, \quad i \in \mathcal{P}, \\ \nabla g_j(x^\star)^\top d &\leq 0, \quad j \in \mathcal{Q}(x^\star), \end{aligned} \tag{1.2}$$

where $\mathcal{Q}(x^\star) = \{j \in \mathcal{Q} \mid g_j(x^\star) = 0\}$ denotes the set of active inequality constraints at $x^\star$.

Proof. Let $d \in \mathbb{R}^n$ be a feasible direction at $x^\star$. By definition, there exists $\bar{\alpha} > 0$ such that

$$x^\star + \alpha d \in \mathcal{F}, \quad \forall \alpha \in (0, \bar{\alpha}].$$

Equality constraints. For each $i \in \mathcal{P}$, feasibility implies

$$h_i(x^\star + \alpha d) = 0 \quad \forall \alpha \in (0, \bar{\alpha}].$$

Using a first-order Taylor expansion of h_i at $x^\star$,

$$h_i(x^\star + \alpha d) = h_i(x^\star) + \alpha \nabla h_i(x^\star)^\top d + o(\alpha).$$

Since $h_i(x^\star) = 0$, dividing by α and letting $\alpha \downarrow 0$ yields

$$\nabla h_i(x^\star)^\top d = 0.$$

Inequality constraints. Consider an inequality constraint indexed by $j \in \mathcal{Q}$. If $j \notin \mathcal{Q}(x^\star)$, then $g_j(x^\star) < 0$ and, by continuity, $g_j(x^\star + \alpha d) < 0$ for all sufficiently small $\alpha > 0$. Hence such constraints impose no restriction on d.

Now let $j \in \mathcal{Q}(x^\star)$, so that $g_j(x^\star) = 0$. Feasibility requires

$$g_j(x^\star + \alpha d) \leq 0 \quad \forall \alpha \in (0, \bar{\alpha}].$$

Using a first-order Taylor expansion,

$$g_j(x^\star + \alpha d) = \alpha \nabla g_j(x^\star)^\top d + o(\alpha).$$

Dividing by α and letting $\alpha \downarrow 0$ gives

$$\nabla g_j(x^\star)^\top d \leq 0.$$

Combining the above conditions proves (1.2). □

The set of feasible directions forms a cone determined entirely by the equality constraints and the active inequality constraints. This theorem shows that local optimality conditions depend only on the active set. If a direction violates none of the linearized active constraints, then it represents an admissible infinitesimal movement of the decision variable. First- and second-order optimality conditions require that the objective function cannot be decreased along any such direction.

2 Method of Lagrange

We begin with constrained optimization problems involving *only equality constraints*. This classical setting leads to the method of Lagrange multipliers and provides the conceptual foundation for first-order optimality conditions in constrained optimization.

2.1 Equality-constrained optimization

Consider the equality-constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \end{aligned} \tag{2.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function and $h_i : \mathbb{R}^n \to \mathbb{R}$ are equality constraint functions indexed by the finite set $\mathcal{P} = \{1, \ldots, p\}$.

The feasible set associated with (2.1) is

$$\mathcal{F} = \{x \in \mathbb{R}^n \mid h_i(x) = 0,\ i \in \mathcal{P}\}.$$

Throughout this section, we assume that f is differentiable and that each constraint function h_i is continuously differentiable in a neighborhood of any point of interest.

2.2 Geometric interpretation

At a feasible point $x \in \mathcal{F}$, the gradients $\nabla h_i(x)$, $i \in \mathcal{P}$, are normal to the constraint manifold defined by the equations $h_i(x) = 0$. A vector $d \in \mathbb{R}^n$ represents a feasible first-order perturbation

at x if and only if it lies in the tangent space of $\mathcal{F}$ at x, which is characterized by

$$\nabla h_i(x)^\top d = 0, \quad i \in \mathcal{P}.$$

If $x^\star$ is a local minimizer, then the directional derivative of the objective function must vanish along all feasible directions. Consequently, the gradient $\nabla f(x^\star)$ must be orthogonal to the tangent space of $\mathcal{F}$ at $x^\star$. Geometrically, this means that $\nabla f(x^\star)$ must lie in the normal space of the constraint manifold, which is spanned by the constraint gradients $\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}$.

2.3 First-order necessary condition

The preceding geometric argument yields the fundamental first-order necessary condition for equality-constrained optimization.

Theorem 2.1 (Euler–Lagrange necessary condition)**.** *Consider problem* (2.1)*. Suppose that f is differentiable and that each constraint function h_i, $i \in \mathcal{P}$, is continuously differentiable at a feasible point $x^\star \in \mathcal{F}$. If $x^\star$ is a local minimizer, then*

$$\nabla f(x^\star) \in \operatorname{span}\{\nabla h_1(x^\star), \ldots, \nabla h_p(x^\star)\}.$$

Equivalently, the vectors

$$\nabla f(x^\star),\ \nabla h_1(x^\star),\ \ldots,\ \nabla h_p(x^\star)$$

are linearly dependent.

Proof. Let $x^\star \in \mathcal{F}$ be a local minimizer of the equality-constrained problem. By definition of local optimality, there exists $\delta > 0$ such that

$$f(x^\star) \le f(x) \quad \text{for all } x \in \mathcal{F} \cap B_\delta(x^\star),$$

where $B_\delta(x^\star)$ denotes the open ball of radius δ centered at $x^\star$.

Let $d \in \mathbb{R}^n$ be an arbitrary feasible first-order perturbation at $x^\star$. By definition, there exists a differentiable curve

$$\gamma : (-\varepsilon, \varepsilon) \to \mathcal{F}$$

for some $\varepsilon > 0$ such that

$$\gamma(0) = x^\star, \qquad \dot\gamma(0) = d.$$

The vector d represents the instantaneous velocity of a feasible curve passing through $x^\star$.

Since $\gamma(t)$ is feasible for all t, each equality constraint is satisfied identically along the curve:

$$h_i(\gamma(t)) = 0, \quad i \in \mathcal{P}, \quad \forall t \in (-\varepsilon, \varepsilon).$$

Differentiating with respect to t at $t = 0$ and applying the chain rule yields

$$\nabla h_i(x^\star)^\top \dot{\gamma}(0) = \nabla h_i(x^\star)^\top d = 0, \quad i \in \mathcal{P}.$$

Thus, every feasible first-order perturbation d satisfies

$$\nabla h_i(x^\star)^\top d = 0, \quad i \in \mathcal{P},$$

which characterizes the tangent space of the feasible set at $x^\star$.

Define the scalar function $\phi(t) := f(\gamma(t))$. Since $\gamma(0) = x^\star$ and $\gamma(t) \in \mathcal{F}$ for all t, the local minimality of $x^\star$ implies that ϕ attains a local minimum at $t = 0$. Consequently, the first derivative of ϕ at $t = 0$ must vanish:

$$\phi'(0) = 0.$$

Applying the chain rule again gives

$$\phi'(0) = \nabla f(x^\star)^\top \dot{\gamma}(0) = \nabla f(x^\star)^\top d.$$

Because d was an arbitrary feasible first-order perturbation, it follows that

$$\nabla f(x^\star)^\top d = 0 \quad \text{for all } d \text{ such that } \nabla h_i(x^\star)^\top d = 0,\ i \in \mathcal{P}.$$

This shows that $\nabla f(x^\star)$ is orthogonal to every vector in the tangent space of the feasible set at $x^\star$. By basic linear algebra, the orthogonal complement of this tangent space is the span of the normal vectors $\{\nabla h_i(x^\star)\}_{i \in \mathcal{P}}$. Therefore,

$$\nabla f(x^\star) \in \operatorname{span}\{\nabla h_1(x^\star), \ldots, \nabla h_p(x^\star)\},$$

which establishes the Euler–Lagrange necessary condition. □

Remark 2.2. For the Euler–Lagrange necessary condition, the differentiability of the objective

and constraint functions at $x^\star$ is sufficient. Continuous differentiability is assumed here to ensure a well-defined local geometry of the feasible set and to align with the regularity requirements of subsequent optimality conditions.

2.4 Lagrangian and Lagrange multipliers

If the constraint gradients $\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}$ are linearly independent, the conclusion of Theorem 2.1 implies the existence of scalars $\lambda_i \in \mathbb{R}$ such that

$$\nabla f(x^\star) + \sum_{i\in\mathcal{P}} \lambda_i \nabla h_i(x^\star) = 0. \tag{2.2}$$

This observation leads directly to the classical method of Lagrange multipliers.

Definition 2.3 (Lagrangian)**.** The *Lagrangian* associated with problem (2.1) is defined as

$$\mathcal{L}(x,\lambda) = f(x) + \sum_{i\in\mathcal{P}} \lambda_i h_i(x),$$

where $\lambda = (\lambda_i)_{i\in\mathcal{P}} \in \mathbb{R}^p$ is the vector of Lagrange multipliers.

The method of Lagrange consists of solving the system

$$\nabla_x \mathcal{L}(x,\lambda) = \nabla f(x) + \sum_{i\in\mathcal{P}} \lambda_i \nabla h_i(x) = 0, \tag{2.3}$$

$$h_i(x) = 0, \quad i \in \mathcal{P}. \tag{2.4}$$

Definition 2.4 (Lagrange candidate)**.** A pair $(x^\star, \lambda^\star)$ satisfying (2.3)–(2.4) is called a *Lagrange candidate*. Any local minimizer at which the constraint gradients are linearly independent must be a Lagrange candidate.

The method of Lagrange multipliers provides *necessary* conditions for local optimality, but not sufficient ones. A Lagrange candidate may correspond to a local maximum or a saddle point rather than a local minimum. Distinguishing among these possibilities requires second-order analysis based on the Hessian of the Lagrangian restricted to feasible directions.

Moreover, if the constraint gradients are linearly dependent at a local minimizer, the classical Lagrange multiplier equation may fail to admit a solution. This degeneracy motivates the introduction of more general first-order conditions, such as the Fritz–John conditions, and the use of constraint qualifications to recover Karush–Kuhn–Tucker optimality conditions in the presence of inequality constraints.

Example 2.5 (Minimization on a nonlinear manifold)**.** Consider the equality-constrained optimization problem

$$\begin{aligned} \min_{(x,y)\in\mathbb{R}^2} \quad & f(x,y) = x^2 + y^2 \\ \text{s.t.} \quad & h(x,y) = x^2 y - 1 = 0. \end{aligned} \tag{2.5}$$

The feasible set is the nonlinear curve

$$\mathcal{F} = \{(x,y) \in \mathbb{R}^2 \mid x^2 y = 1\},$$

which consists of two symmetric branches corresponding to $x > 0$ and $x < 0$.

The associated Lagrangian is

$$\mathcal{L}(x,y,\lambda) = x^2 + y^2 + \lambda(x^2 y - 1).$$

The first-order stationarity conditions are obtained by setting the partial derivatives of $\mathcal{L}$ with respect to x and y equal to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x + 2\lambda x y = 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = 2y + \lambda x^2 = 0,$$

together with the feasibility condition $x^2 y = 1$.

Since $x = 0$ is infeasible, the first equation implies

$$1 + \lambda y = 0 \quad \Rightarrow \quad \lambda = -\frac{1}{y}.$$

Substituting this expression into the second stationarity condition yields

$$2y - \frac{x^2}{y} = 0 \quad \Rightarrow \quad 2y^2 = x^2.$$

Using the constraint $x^2 y = 1$, we obtain

$$x^2 = 2^{1/3}, \qquad y = 2^{-1/3}.$$

Thus, the Lagrange candidates are

$$(x^\star, y^\star) = (\pm 2^{1/6}, 2^{-1/3}), \qquad \lambda^\star = -2^{1/3}.$$

By symmetry of the objective function and the constraint, both points correspond to local minimizers of the constrained problem. Verifying this rigorously requires second-order analysis of the Lagrangian restricted to feasible directions, illustrating the fact that first-order conditions alone are insufficient to fully characterize optimality.

2.5 Second-Order Sufficient Conditions

The first-order Euler–Lagrange condition characterizes candidate solutions by requiring stationarity of the objective function along feasible directions. However, as in unconstrained optimization, first-order conditions alone do not distinguish local minima from local maxima or saddle points. This distinction requires second-order information.

Throughout this subsection, we assume that the objective function f and the constraint functions h_i, $i \in \mathcal{P}$, are *twice continuously differentiable* in a neighborhood of a feasible point $x^\star$.

2.5.1 Tangent space and second-order variations

Let $x^\star \in \mathcal{F}$ be a feasible point at which the gradients $\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}$ are linearly independent. The tangent space of the feasible set at $x^\star$ is given by

$$\mathcal{T}(x^\star) = \left\{d \in \mathbb{R}^n \mid \nabla h_i(x^\star)^\top d = 0,\ i \in \mathcal{P}\right\}.$$

Let $N \in \mathbb{R}^{n\times(n-p)}$ be a matrix whose columns form a basis of $\mathcal{T}(x^\star)$. Any feasible first-order perturbation can be written as $d = N\xi$ for some $\xi \in \mathbb{R}^{n-p}$.

2.5.2 Lagrangian and second-order expansion

If $x^\star$ satisfies the Euler–Lagrange condition, then there exist multipliers $\lambda_i \in \mathbb{R}$, $i \in \mathcal{P}$, such that

$$\nabla f(x^\star) + \sum_{i\in\mathcal{P}} \lambda_i \nabla h_i(x^\star) = 0. \tag{2.6}$$

Define the *Lagrangian* associated with the equality-constrained problem by

$$\mathcal{L}(x,\lambda) = f(x) + \sum_{i\in\mathcal{P}} \lambda_i h_i(x).$$

The second-order behavior of the objective along feasible directions is governed by the Hes-

sian of the Lagrangian,

$$\nabla^2_{xx}\mathcal{L}(x^\star, \lambda) = \nabla^2 f(x^\star) + \sum_{i\in\mathcal{P}} \lambda_i \nabla^2 h_i(x^\star).$$

2.5.3 Second-order sufficient condition

We are now ready to state the second-order sufficient condition for local optimality.

Theorem 2.6 (Second-Order Sufficient Condition for Equality Constraints)**.** *Consider the equality-constrained optimization problem* (2.1)*. Suppose that:*

1. *the objective function f and the constraint functions h_i, $i \in \mathcal{P}$, are twice continuously differentiable in a neighborhood of a feasible point $x^\star$,*
2. *the gradients $\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}$ are linearly independent,*
3. *there exist multipliers $\lambda_i \in \mathbb{R}$ such that the stationarity condition*

$$\nabla f(x^\star) + \sum_{i\in\mathcal{P}} \lambda_i \nabla h_i(x^\star) = 0 \tag{2.7}$$

   holds,

4. *the reduced Hessian*

$$N^\top \left[\nabla^2 f(x^\star) + \sum_{i\in\mathcal{P}} \lambda_i \nabla^2 h_i(x^\star)\right] N \tag{2.8}$$

   is positive definite.

Then $x^\star$ is an isolated local minimum *of the equality-constrained problem.*

Proof. Let $d \in \mathcal{T}(x^\star)$ be an arbitrary nonzero feasible first-order perturbation. By definition of the tangent space, d satisfies

$$\nabla h_i(x^\star)^\top d = 0, \quad i \in \mathcal{P}.$$

Because the gradients of the constraint functions are linearly independent and the constraints are continuously differentiable, the feasible set $\mathcal{F}$ is locally a smooth manifold. Consequently, for every tangent vector $d \in \mathcal{T}(x^\star)$, there exists a continuously differentiable curve

$$\gamma : (-\varepsilon, \varepsilon) \to \mathcal{F}$$

such that

$$\gamma(0) = x^\star, \qquad \dot{\gamma}(0) = d.$$

This curve represents a feasible trajectory passing through $x^\star$ with initial velocity d.

Define the scalar function $\phi(t) = f(\gamma(t))$. Since $x^\star$ satisfies the first-order stationarity condition and $\gamma(t)$ remains feasible for all t, the first derivative of ϕ at $t = 0$ vanishes:

$$\phi'(0) = \nabla f(x^\star)^\top \dot{\gamma}(0) = \nabla f(x^\star)^\top d = 0.$$

Because f and the constraint functions are twice continuously differentiable, a second-order Taylor expansion of ϕ about $t = 0$ yields

$$\phi(t) = \phi(0) + \frac{t^2}{2}\, d^\top \nabla^2_{xx}\mathcal{L}(x^\star, \lambda)\, d + o(t^2),$$

where

$$\nabla^2_{xx}\mathcal{L}(x^\star, \lambda) = \nabla^2 f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i \nabla^2 h_i(x^\star)$$

is the Hessian of the Lagrangian with respect to x.

Since $d \in \mathcal{T}(x^\star)$, there exists a nonzero vector ξ such that $d = N\xi$. Substituting into the quadratic form gives

$$d^\top \nabla^2_{xx}\mathcal{L}(x^\star, \lambda) d = \xi^\top \left(N^\top \nabla^2_{xx}\mathcal{L}(x^\star, \lambda) N \right) \xi.$$

By assumption, the reduced Hessian matrix in (2.8) is positive definite, and therefore

$$d^\top \nabla^2_{xx}\mathcal{L}(x^\star, \lambda) d > 0 \quad \text{for all } d \in \mathcal{T}(x^\star) \setminus \{0\}.$$

It follows that for all sufficiently small $t \neq 0$,

$$f(\gamma(t)) > f(x^\star).$$

Thus, $x^\star$ is a strict local minimum along every feasible curve emanating from $x^\star$. Since every feasible nearby point can be reached (to first order) by such curves, $x^\star$ is an isolated local minimum of the equality-constrained problem. □

Remark 2.7. The matrix in (2.8) is often referred to as the *reduced Hessian*. Its positive definiteness ensures that the second-order variation of the objective is strictly positive along all feasible directions. This condition generalizes the classical unconstrained second-order sufficient condition to the constrained setting.

Example 2.8 (Second-order sufficiency for an equality-constrained problem)**.** Consider the equality-constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & f(x) = x_1^2 + x_2^2 + x_3^2 \\ \text{s.t.} \quad & h(x) := x_1 + x_2 + x_3 - 1 = 0. \end{aligned} \tag{2.9}$$

The feasible set is the affine plane

$$\mathcal{F} = \{\, x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 1 \,\}.$$

First-order optimality. The gradients of the objective and constraint functions are

$$\nabla f(x) = (2x_1,\, 2x_2,\, 2x_3)^\top, \qquad \nabla h(x) = (1,\, 1,\, 1)^\top.$$

The Euler–Lagrange (Lagrange multiplier) condition requires the existence of $\lambda \in \mathbb{R}$ such that

$$\nabla f(x^\star) + \lambda \nabla h(x^\star) = 0.$$

This yields the system

$$(2x_1^\star,\, 2x_2^\star,\, 2x_3^\star)^\top + \lambda(1,1,1)^\top = 0,$$

from which it follows immediately that

$$x_1^\star = x_2^\star = x_3^\star = \tfrac{1}{3}, \qquad \lambda^\star = -\tfrac{2}{3}.$$

Thus,

$$x^\star = \left(\tfrac{1}{3},\, \tfrac{1}{3},\, \tfrac{1}{3}\right)$$

is a stationary feasible point.

Tangent space. Since

$$\nabla h(x^\star) = (1,1,1)^\top,$$

a vector $d = (d_1, d_2, d_3)^\top$ is a feasible first-order perturbation if and only if

$$\nabla h(x^\star)^\top d = d_1 + d_2 + d_3 = 0.$$

Hence, the tangent space of the feasible set at $x^\star$ is the two-dimensional subspace

$$\mathcal{T}(x^\star) = \left\{ d \in \mathbb{R}^3 \mid d_1 + d_2 + d_3 = 0 \right\}.$$

To construct a basis, parameterize all solutions of $d_1 + d_2 + d_3 = 0$. Let $d_1 = \alpha$ and $d_2 = \beta$ be free parameters. Then

$$d_3 = -\alpha - \beta,$$

and every tangent direction can be written as

$$d = (\alpha, \beta, -\alpha - \beta)^\top = \alpha(1, 0, -1)^\top + \beta(0, 1, -1)^\top.$$

Define

$$n_1 := (1, 0, -1)^\top, \qquad n_2 := (0, 1, -1)^\top, \qquad N := [\, n_1 \;\; n_2 \,] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}.$$

Each column of N lies in $\mathcal{T}(x^\star)$ and the two columns are linearly independent. Therefore, they form a basis of the tangent space. Moreover, every $d \in \mathcal{T}(x^\star)$ admits a unique representation $d = Nv$ for some $v \in \mathbb{R}^2$.

Second-order condition. The Hessians of the objective and constraint functions are

$$\nabla^2 f(x) = 2I_3, \qquad \nabla^2 h(x) = 0.$$

Consequently, the Hessian of the Lagrangian at $(x^\star, \lambda^\star)$ is

$$\nabla^2_{xx} \mathcal{L}(x^\star, \lambda^\star) = 2I_3.$$

The reduced Hessian is given by

$$N^\top \nabla^2_{xx} \mathcal{L}(x^\star, \lambda^\star) N = 2N^\top N = 2 \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The eigenvalues of this matrix are 2 and 6, both strictly positive. Hence, the reduced Hessian is positive definite on the tangent space.

All assumptions of the second-order sufficient condition for equality-constrained problems are satisfied. Therefore,

$$x^\star = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right)$$

is an *isolated local minimum* of the equality-constrained problem.

3 Fritz–John and Mangasarian–Fromovitz Conditions

Fritz–John conditions (1948). The earliest general first-order necessary conditions for constrained optimization were developed by Fritz John in 1948 [6]. John generalized the classical method of Lagrange multipliers to optimization problems involving both equality and inequality constraints, without imposing any regularity assumptions on the constraint set. The resulting Fritz–John conditions assert that, at a local minimum, there exist multipliers associated with the objective function and the constraints—not all zero—that satisfy a stationarity condition together with complementary slackness.

A distinctive feature of John's formulation is the presence of an additional multiplier associated with the objective function, which is allowed to vanish. While this makes the Fritz–John conditions universally applicable, it also allows degenerate situations in which the objective function does not appear in the stationarity condition. Such cases later became known as *abnormal multipliers* and motivated the development of constraint qualifications.

Karush–Kuhn–Tucker conditions (1951). The conditions now known as the Karush–Kuhn–Tucker (KKT) conditions emerged in the early development of nonlinear programming. An early version appeared in the 1939 master's thesis of William Karush [8], but the framework was independently and systematically developed by Kuhn and Tucker in 1951 [9], motivated in part by applications to economics and game theory.

The KKT conditions resemble the Fritz–John conditions but impose a normalization that fixes the multiplier associated with the objective function to one. This normalization eliminates abnormal multipliers and yields a clean stationarity condition involving only the gradients of the objective function and the active constraints. However, the validity of this normalization depends on additional regularity assumptions on the constraint set, which were not formalized at the time.

Mangasarian–Fromovitz constraint qualification (1967). The need to precisely characterize when the KKT conditions are valid led to the systematic study of constraint qualifications in the 1960s. A particularly influential contribution was made by Mangasarian and Fromovitz in 1967 [11], who introduced what is now known as the Mangasarian–Fromovitz Constraint Qualification (MFCQ).

MFCQ requires linear independence of the equality constraint gradients and the existence of a feasible direction that strictly decreases all active inequality constraints. Mangasarian and Fromovitz showed that this condition rules out abnormal Fritz–John multipliers and guarantees that the multiplier associated with the objective function is strictly positive. Under MFCQ, the Fritz–John conditions reduce exactly to the Karush–Kuhn–Tucker conditions.

From a modern perspective, these developments establish a clear logical hierarchy in first-order optimality theory. The Fritz–John conditions provide the most general necessary conditions, valid without regularity assumptions. Constraint qualifications such as MFCQ identify settings in which degeneracy is excluded. Under such conditions, the Fritz–John conditions reduce to the KKT conditions, which form the cornerstone of modern nonlinear programming [3, 4, 12].

3.1 Theorems of Alternatives: Gordan, Farkas, and Motzkin

We now present a family of fundamental results from linear algebra and convex analysis known as *theorems of alternatives*. These results assert that, given a pair of linear systems of a certain form, *exactly one* of the two systems admits a solution. Such results formalize a basic geometric dichotomy: either a feasible direction exists, or there is a dual certificate proving that no such direction can exist.

The classical theorems of alternatives form a strict hierarchy in expressive power:

$$\text{Motzkin's Transposition Theorem} \supset \text{Farkas' Lemma} \supset \text{Gordan's Theorem.}$$

All three results are equivalent in finite dimensions, but they differ in how they treat equalities, weak inequalities, and strict inequalities. This distinction explains why different theorems appear naturally in different parts of constrained optimization theory.

3.1.1 Gordan's theorem

We begin with the most restrictive but conceptually transparent result.

Theorem 3.1 (Gordan's Theorem of the Alternative)**.** *Let $B \in \mathbb{R}^{q \times n}$. Exactly one of the following two systems has a solution:*

1. *There exists $d \in \mathbb{R}^n$ such that*

$$Bd < 0.$$

2. *There exists $\mu \in \mathbb{R}^q_{\geq 0}$, $\mu \neq 0$, such that*

$$B^\top \mu = 0.$$

The two systems are mutually exclusive.

Proof. We show that exactly one of the two systems can hold.

Assume first that there exists $d \in \mathbb{R}^n$ such that $Bd < 0$. Suppose, for contradiction, that there also exists $\mu \in \mathbb{R}^q_{\geq 0}$, $\mu \neq 0$, such that $B^\top \mu = 0$. Taking the inner product with d yields

$$0 = \mu^\top (Bd) < 0,$$

since $\mu \geq 0$, $\mu \neq 0$, and $Bd < 0$, which is a contradiction. Hence, the second system cannot hold if the first does.

Conversely, assume that there is no $d \in \mathbb{R}^n$ such that $Bd < 0$. Then the system $Bd \leq 0$ admits only the trivial solution $d = 0$. By the separating hyperplane theorem, the cone

$$\{Bd \mid d \in \mathbb{R}^n\}$$

can be separated from the negative orthant. This implies the existence of a nonzero vector $\mu \in \mathbb{R}^q_{\geq 0}$ such that

$$B^\top \mu = 0.$$

Thus, exactly one of the two systems is solvable. □

Gordan's theorem concerns *homogeneous* systems and *strict* inequalities. Geometrically, it asserts that either there exists a direction that strictly decreases all components of Bd, or there exists a nontrivial nonnegative linear combination of the rows of B that annihilates all such directions. This theorem is the analytical backbone of strict feasibility conditions and appears naturally in constraint qualifications such as the Mangasarian–Fromovitz condition.

Example 3.2 (Illustration of Gordan's Theorem)**.** We present two simple examples showing that exactly one of the two systems in Theorem 3.1 holds.

Case 1: Existence of a strictly feasible direction. Let

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 2}.$$

Consider the vector

$$d = \begin{pmatrix} -1 \\ -1 \end{pmatrix}.$$

Then

$$Bd = \begin{pmatrix} -1 \\ -1 \end{pmatrix} < 0,$$

so system (1) of Theorem 3.1 is satisfied.

We now check system (2). The condition $B^\top \mu = 0$ becomes

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \qquad \mu_1 \geq 0,\ \mu_2 \geq 0, \qquad \mu_1 = 0,\ \mu_2 = 0.$$

The only solution is $\mu = 0$, which is excluded. Hence system (2) has no solution, as predicted by Gordan's theorem.

Case 2: Existence of a nontrivial dual certificate. Let

$$B = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 2}.$$

For any $d = (d_1, d_2)^\top \in \mathbb{R}^2$, we have

$$Bd = \begin{pmatrix} d_1 \\ -d_1 \end{pmatrix}.$$

It is impossible for both components to be strictly negative, so there exists no d such that $Bd < 0$. Thus system (1) has no solution.

Now consider system (2). Choose

$$\mu = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \in \mathbb{R}^2_{\geq 0}, \qquad \mu \neq 0.$$

Then

$$B^\top \mu = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus system (2) is satisfied.

In the first case, there exists a direction that strictly decreases all linear forms defined by the rows of B. In the second case, no such direction exists, and this infeasibility is certified by a nontrivial nonnegative combination of the rows of B that annihilates all directions.

3.1.2 Farkas' Lemma

Farkas' Lemma generalizes Gordan's theorem by allowing equality constraints and weak inequalities. This added flexibility makes it the central tool for first-order optimality conditions.

Theorem 3.3 (Farkas' Lemma (Conic Form))**.** *Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{q\times n}$, and $w \in \mathbb{R}^n$ be given. Exactly one of the following two systems has a solution:*

1. *There exists $d \in \mathbb{R}^n$ such that*

$$Ad = 0, \qquad Bd \leq 0, \qquad w^\top d < 0. \tag{3.1}$$

2. *There exist multipliers $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}^q_{\geq 0}$ such that*

$$w = A^\top \lambda + B^\top \mu. \tag{3.2}$$

Moreover, the two systems are mutually exclusive.

Proof. Define the set

$$\mathcal{D} = \{d \in \mathbb{R}^n \mid Ad = 0,\ Bd \leq 0\}.$$

The set $\mathcal{D}$ is a closed convex cone.

Assume that there exist $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}^q_{\geq 0}$ such that $w = A^\top \lambda + B^\top \mu$. Then for any $d \in \mathcal{D}$,

$$w^\top d = \lambda^\top (Ad) + \mu^\top (Bd) = 0 + \mu^\top (Bd) \geq 0,$$

which shows that no $d \in \mathcal{D}$ can satisfy $w^\top d < 0$. Hence, the first system is infeasible.

Conversely, assume that no $d \in \mathcal{D}$ satisfies $w^\top d < 0$. Then

$$w^\top d \geq 0 \quad \forall d \in \mathcal{D},$$

so w belongs to the dual cone

$$\mathcal{D}^* = \{y \in \mathbb{R}^n \mid y^\top d \geq 0 \, \forall d \in \mathcal{D}\}.$$

Standard results in convex analysis imply that

$$\mathcal{D}^* = \operatorname{span}\{A^\top\} + \operatorname{cone}\{B^\top\}.$$

Therefore, there exist $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}^q_{\geq 0}$ such that

$$w = A^\top \lambda + B^\top \mu,$$

which establishes the second system. Mutual exclusivity follows from the above arguments. □

Farkas' Lemma states that either there exists a direction satisfying the linear constraints and strictly decreasing the linear functional $w^\top d$, or w can be expressed as a linear combination of the constraint gradients with nonnegative multipliers. This result is the precise mathematical mechanism that converts the absence of feasible descent directions into the existence of Lagrange multipliers. It underlies the Fritz–John conditions and the Karush–Kuhn–Tucker conditions.

If the equality constraints $Ad = 0$ are removed and the inequality $Bd \leq 0$ is strengthened to $Bd < 0$, Farkas' Lemma reduces to Gordan's theorem.

3.1.3 Motzkin's transposition theorem

The most general theorem of alternatives is due to Motzkin and allows both weak and strict inequalities in the same system.

Theorem 3.4 (Motzkin's Transposition Theorem)**.** *Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{q \times n}$, $C \in \mathbb{R}^{r \times n}$, and $b \in \mathbb{R}^m$, $c \in \mathbb{R}^r$ be given. Exactly one of the following systems has a solution:*

1. *There exists $x \in \mathbb{R}^n$ such that*

$$Ax \leq b, \qquad Cx < c.$$

2. *There exist multipliers $y \in \mathbb{R}^m_{\geq 0}$ and $z \in \mathbb{R}^r_{\geq 0}$, not both zero, such that*

$$A^\top y + C^\top z = 0, \qquad b^\top y + c^\top z < 0.$$

Proof. Motzkin's theorem follows from a separation argument in an augmented space.

Assume that there exists $x \in \mathbb{R}^n$ such that

$$Ax \leq b, \qquad Cx < c.$$

Suppose, for contradiction, that there also exist $y \in \mathbb{R}^m_{\geq 0}$ and $z \in \mathbb{R}^r_{\geq 0}$, not both zero, such that

$$A^\top y + C^\top z = 0, \qquad b^\top y + c^\top z < 0.$$

Taking the inner product with x yields

$$0 = y^\top (Ax) + z^\top (Cx) \leq y^\top b + z^\top c < 0,$$

which is a contradiction.

Conversely, assume that the system $Ax \leq b, Cx < c$ has no solution. Define the sets

$$\mathcal{S}_1 = \{(Ax, Cx) \mid x \in \mathbb{R}^n\}, \qquad \mathcal{S}_2 = \{(u, v) \mid u \leq b,\ v < c\}.$$

These sets are convex and disjoint. By the separating hyperplane theorem, there exists a nonzero vector (y, z) with $y \geq 0, z \geq 0$ such that

$$y^\top A + z^\top C = 0, \qquad b^\top y + c^\top z < 0.$$

This yields the second system. The two systems are mutually exclusive by construction. □

Motzkin's theorem strictly generalizes Farkas' Lemma by permitting strict and weak inequalities to coexist. Farkas' Lemma is obtained as a special case by eliminating strict inequalities, while Gordan's theorem arises as a homogeneous specialization. Although Motzkin's theorem is rarely stated explicitly in basic optimization texts, it provides the most general sep-

aration principle underlying all linear constraint systems.

Gordan's theorem captures strict feasibility and appears in constraint qualifications such as the Mangasarian–Fromovitz condition. Farkas' Lemma converts the nonexistence of feasible descent directions into the existence of Lagrange multipliers and is central to KKT theory. Motzkin's theorem provides the most general alternative, unifying strict and weak feasibility in a single framework. These results provide the algebraic foundation for modern optimality theory.

3.2 First-Order Necessary Conditions for General Constrained Programs

We now state a general first-order necessary condition for optimality for constrained optimization problems of the form (1.1). This condition applies to problems with both equality and inequality constraints and requires only first-order differentiability.

Theorem 3.5 (First-Order Necessary Condition (Fritz–John / Mangasarian–Fromovitz))**.** *Consider the constrained optimization problem* (1.1)*. Assume that the objective function f and the inequality constraint functions g_j, $j \in \mathcal{Q}$, are differentiable at a feasible point $x^\star \in \mathcal{F}$, and that the equality constraint functions h_i, $i \in \mathcal{P}$, are continuously differentiable at $x^\star$.*

If $x^\star$ is a local minimum, then there exist scalars

$$u_0 \geq 0, \qquad u_j \geq 0 \ \ (j \in \mathcal{Q}), \qquad \lambda_i \in \mathbb{R} \ \ (i \in \mathcal{P}),$$

not all zero, such that

$$u_0 \nabla f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}} u_j \nabla g_j(x^\star) = 0, \tag{3.3}$$

together with the complementary slackness conditions

$$u_j g_j(x^\star) = 0, \quad j \in \mathcal{Q}. \tag{3.4}$$

Proof. The proof is based on feasible directions and a theorem of alternatives.

Let $x^\star \in \mathcal{F}$ be a local minimum of problem (1.1). By definition, there exists a neighborhood $N(x^\star)$ such that

$$f(x^\star) \leq f(x), \quad \forall x \in \mathcal{F} \cap N(x^\star).$$

Step 1: Absence of feasible descent directions. Let $d \in \mathbb{R}^n$ be any feasible direction at $x^\star$, i.e., $x^\star + \alpha d \in \mathcal{F}$ for all sufficiently small $\alpha > 0$. Define $\phi(\alpha) = f(x^\star + \alpha d)$. Since $x^\star$ is a local minimum, $\phi(\alpha) \geq \phi(0)$ for all sufficiently small $\alpha > 0$. Differentiating at $\alpha = 0^+$ yields

$$\nabla f(x^\star)^\top d = \phi'(0^+) \geq 0.$$

Hence,

$$\nabla f(x^\star)^\top d \geq 0 \quad \text{for all feasible directions } d. \tag{3.5}$$

Step 2: Linearized description of feasible directions. By Theorem 1.7, every feasible direction d satisfies

$$\nabla h_i(x^\star)^\top d = 0, \quad i \in \mathcal{P},$$
$$\nabla g_j(x^\star)^\top d \leq 0, \quad j \in \mathcal{Q}(x^\star),$$

where $\mathcal{Q}(x^\star) = \{j \in \mathcal{Q} \mid g_j(x^\star) = 0\}$ is the active set. Consequently, inequality (3.5) holds for all d in the closed convex cone

$$\mathcal{D} = \left\{d \in \mathbb{R}^n \mid \nabla h_i(x^\star)^\top d = 0,\ i \in \mathcal{P},\ \nabla g_j(x^\star)^\top d \leq 0,\ j \in \mathcal{Q}(x^\star)\right\}.$$

Step 3: Application of Farkas' Lemma. Condition (3.5) states that the linear functional $d \mapsto \nabla f(x^\star)^\top d$ is nonnegative over the cone $\mathcal{D}$. Equivalently, there exists no $d \in \mathcal{D}$ such that $\nabla f(x^\star)^\top d < 0$.

By Farkas' Lemma (Theorem 3.3), exactly one of the following holds:

- there exists $d \in \mathcal{D}$ such that $\nabla f(x^\star)^\top d < 0$, or
- there exist multipliers

$$u_0 \geq 0, \quad \lambda_i \in \mathbb{R}\ (i \in \mathcal{P}), \quad u_j \geq 0\ (j \in \mathcal{Q}(x^\star)),$$

  not all zero, such that

$$u_0 \nabla f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}(x^\star)} u_j \nabla g_j(x^\star) = 0. \tag{3.6}$$

Since the first alternative is ruled out by (3.5), the second must hold, yielding the stationarity condition.

Step 4: Complementary slackness. For any $j \notin \mathcal{Q}(x^\star)$, we have $g_j(x^\star) < 0$. Such constraints do not restrict feasible directions and therefore cannot appear in the conic representation (3.6). Hence,

$$u_j = 0, \quad j \notin \mathcal{Q}(x^\star),$$

which implies the complementary slackness conditions

$$u_j g_j(x^\star) = 0, \quad j \in \mathcal{Q}.$$

The existence of multipliers satisfying (3.6) together with complementary slackness establishes the Fritz–John optimality conditions (3.3)–(3.4), completing the proof. □

Definition 3.6 (Mangasarian–Fromovitz Constraint Qualification (MFCQ))**.** Consider the constrained optimization problem (1.1), and let $x^\star \in \mathcal{F}$ be a feasible point. Denote the active set of inequality constraints at $x^\star$ by

$$\mathcal{Q}(x^\star) = \{\, j \in \mathcal{Q} \mid g_j(x^\star) = 0 \,\}.$$

The Mangasarian–Fromovitz Constraint Qualification (MFCQ) is said to hold at $x^\star$ if the following conditions are satisfied:

1. The gradients of the equality constraints are linearly independent at $x^\star$, i.e.,

$$\{\nabla h_i(x^\star)\}_{i \in \mathcal{P}} \quad \text{are linearly independent.}$$

2. There exists a vector $d \in \mathbb{R}^n$ such that

$$\nabla h_i(x^\star)^\top d = 0, \quad i \in \mathcal{P}, \qquad \nabla g_j(x^\star)^\top d < 0, \quad j \in \mathcal{Q}(x^\star).$$

Theorem 3.7 (Karush–Kuhn–Tucker Theorem under the Mangasarian–Fromovitz Constraint Qualification)**.** *Consider the constrained optimization problem* (1.1)*. Assume that the objective function f and the constraint functions h_i, $i \in \mathcal{P}$, and g_j, $j \in \mathcal{Q}$, are continuously differentiable at a feasible point $x^\star \in \mathcal{F}$.*

If $x^\star$ is a local minimum and the Mangasarian–Fromovitz Constraint Qualification (MFCQ) holds

at $x^\star$, then there exist Lagrange multipliers

$$\lambda_i \in \mathbb{R}, \quad i \in \mathcal{P}, \qquad \mu_j \geq 0, \quad j \in \mathcal{Q},$$

such that the following Karush–Kuhn–Tucker conditions are satisfied:

$$\nabla f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}} \mu_j \nabla g_j(x^\star) = 0, \tag{3.7}$$

$$h_i(x^\star) = 0, \quad i \in \mathcal{P}, \tag{3.8}$$

$$g_j(x^\star) \leq 0, \quad j \in \mathcal{Q}, \tag{3.9}$$

$$\mu_j g_j(x^\star) = 0, \quad j \in \mathcal{Q}. \tag{3.10}$$

Proof. Since $x^\star$ is a local minimum and all functions are continuously differentiable at $x^\star$, the Fritz–John conditions (Theorem 3.5) apply. Hence, there exist multipliers

$$u_0 \geq 0, \quad \lambda_i \in \mathbb{R},\ i \in \mathcal{P}, \quad u_j \geq 0,\ j \in \mathcal{Q},$$

not all zero, such that

$$u_0 \nabla f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}} u_j \nabla g_j(x^\star) = 0, \tag{3.11}$$

together with the complementary slackness conditions

$$u_j g_j(x^\star) = 0, \quad j \in \mathcal{Q}.$$

Step 1: Reduction to active constraints. By complementary slackness, $u_j = 0$ for all $j \notin \mathcal{Q}(x^\star)$, where

$$\mathcal{Q}(x^\star) = \{j \in \mathcal{Q} \mid g_j(x^\star) = 0\}$$

denotes the active set. Thus (3.11) may be written as

$$u_0 \nabla f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}(x^\star)} u_j \nabla g_j(x^\star) = 0. \tag{3.12}$$

Step 2: Exclusion of the abnormal case $u_0 = 0$. Assume, for contradiction, that $u_0 = 0$. Then (3.12) reduces to

$$\sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}(x^\star)} u_j \nabla g_j(x^\star) = 0, \qquad u_j \geq 0. \tag{3.13}$$

Since the gradients $\{\nabla h_i(x^\star)\}_{i \in \mathcal{P}}$ are linearly independent (by MFCQ), and MFCQ guarantees the existence of a direction $d \in \mathbb{R}^n$ such that

$$\nabla h_i(x^\star)^\top d = 0, \quad i \in \mathcal{P}, \qquad \nabla g_j(x^\star)^\top d < 0, \quad j \in \mathcal{Q}(x^\star),$$

we may take the inner product of (3.13) with d to obtain

$$0 = \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star)^\top d + \sum_{j \in \mathcal{Q}(x^\star)} u_j \nabla g_j(x^\star)^\top d = \sum_{j \in \mathcal{Q}(x^\star)} u_j \nabla g_j(x^\star)^\top d.$$

Since $\nabla g_j(x^\star)^\top d < 0$ for all $j \in \mathcal{Q}(x^\star)$ and $u_j \geq 0$, the above equality can hold only if

$$u_j = 0 \quad \forall j \in \mathcal{Q}(x^\star).$$

Substituting into (3.13) yields

$$\sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) = 0.$$

By linear independence of the equality constraint gradients, this implies $\lambda_i = 0$ for all $i \in \mathcal{P}$.

Thus all multipliers vanish, contradicting the Fritz–John condition that the multipliers are not all zero. Hence $u_0 > 0$.

Step 3: Normalization and KKT conditions. Since $u_0 > 0$, we may normalize the Fritz–John multipliers by dividing (3.12) by u_0 and define

$$\mu_j = \frac{u_j}{u_0}, \quad j \in \mathcal{Q}, \qquad \lambda_i = \frac{\lambda_i}{u_0}, \quad i \in \mathcal{P}.$$

This yields the stationarity condition

$$\nabla f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}} \mu_j \nabla g_j(x^\star) = 0,$$

together with feasibility and complementary slackness,

$$h_i(x^\star) = 0, \quad g_j(x^\star) \le 0, \quad \mu_j g_j(x^\star) = 0, \quad j \in \mathcal{Q}.$$

These are precisely the Karush–Kuhn–Tucker conditions.

Under the Mangasarian–Fromovitz Constraint Qualification, every local minimum satisfies the Karush–Kuhn–Tucker conditions. □

Example 3.8 (Nonlinear degeneracy and failure of MFCQ)**.** Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & f(x) = x_3 \\ \text{s.t.} \quad & h(x) := x_1^2 + x_2^2 - x_3^2 = 0, \\ & g_1(x) := x_3 \le 0, \\ & g_2(x) := x_1 + x_3 \le 0, \\ & g_3(x) := x_2 + x_3 \le 0. \end{aligned} \tag{3.14}$$

The equality constraint defines a double cone

$$x_3^2 = x_1^2 + x_2^2,$$

while the inequality constraint $x_3 \le 0$ restricts attention to the lower nappe. The remaining inequalities further intersect the cone with tilted half-spaces. The feasible set is therefore a curved cone with a degenerate apex at the origin.

The gradients of the constraint functions are

$$\nabla h(x) = (2x_1, 2x_2, -2x_3), \quad \nabla g_1 = (0,0,1), \quad \nabla g_2 = (1,0,1), \quad \nabla g_3 = (0,1,1).$$

A regular boundary point. Consider the feasible point

$$x^{(1)} = (-1, -1, -\sqrt{2}).$$

At this point, the equality constraint is satisfied and only the inequality g_1 is active. Thus,

$$\mathcal{Q}(x^{(1)}) = \{1\}.$$

The Mangasarian–Fromovitz Constraint Qualification requires the existence of a direction $d \in \mathbb{R}^3$ such that

$$\nabla h(x^{(1)})^\top d = 0, \qquad \nabla g_1(x^{(1)})^\top d < 0.$$

Choosing, for instance,

$$d = (1, -1, -\sqrt{2}),$$

we obtain

$$\nabla h(x^{(1)})^\top d = 0, \qquad \nabla g_1(x^{(1)})^\top d = -\sqrt{2} < 0.$$

Hence, MFCQ holds at $x^{(1)}$. If $x^{(1)}$ were a local minimum, the Karush–Kuhn–Tucker conditions would necessarily hold at this point.

A degenerate point where MFCQ fails. Now consider the point

$$x^{(2)} = (0, 0, 0).$$

This point satisfies all constraints, and all inequality constraints are active:

$$\mathcal{Q}(x^{(2)}) = \{1, 2, 3\}.$$

The gradients at $x^{(2)}$ are

$$\nabla h(x^{(2)}) = (0, 0, 0), \quad \nabla g_1 = (0, 0, 1), \quad \nabla g_2 = (1, 0, 1), \quad \nabla g_3 = (0, 1, 1).$$

The MFCQ condition would require a direction $d \in \mathbb{R}^3$ satisfying

$$\nabla g_1^\top d < 0, \qquad \nabla g_2^\top d < 0, \qquad \nabla g_3^\top d < 0,$$

that is,

$$d_3 < 0, \qquad d_1 + d_3 < 0, \qquad d_2 + d_3 < 0.$$

However, at the apex of the cone all feasible first-order directions are tangent to the equality constraint, and no strictly inward direction exists that simultaneously satisfies all three inequalities. Consequently, no such d exists, and MFCQ fails at $x^{(2)}$.

This example illustrates that the failure of the Mangasarian–Fromovitz Constraint Qualification is not caused by algebraic dependence of constraint gradients, but by geometric degen-

eracy of the feasible set. At $x^{(2)}$, the feasible region is "pinched" at the cone apex, eliminating all strictly feasible first-order directions. At such points, the Karush–Kuhn–Tucker conditions are not guaranteed to hold, and only the weaker Fritz–John necessary conditions apply.

3.3 Second-Order Conditions for General Constrained Programs

We now turn to second-order optimality conditions for constrained optimization problems of the form (1.1), allowing both equality and inequality constraints. In contrast to the equality-constrained case, second-order conditions for general nonlinear programs must account for the interaction between curvature, active constraints, and the geometry of feasible directions.

Throughout this subsection, we assume that the objective function f and all constraint functions h_i, $i \in \mathcal{P}$, and g_j, $j \in \mathcal{Q}$ are twice continuously differentiable in a neighborhood of a feasible point $x^\star \in \mathcal{F}$.

Second-order behavior is governed by directions that are feasible to first order and along which the objective does not decrease at first order.

Definition 3.9 (Critical direction set)**.** The set of *critical directions* at $x^\star$ is

$$D(x^\star) = \left\{ d \in \mathbb{R}^n \;\middle|\; \begin{array}{ll} \nabla f(x^\star)^\top d \le 0, & \\ \nabla h_i(x^\star)^\top d = 0, & i \in \mathcal{P}, \\ \nabla g_j(x^\star)^\top d \le 0, & j \in \mathcal{Q}(x^\star) \end{array} \right\}. \tag{3.15}$$

The set $D(x^\star)$ is a closed convex cone and is always nonempty (it contains $d = 0$). Intuitively, $D(x^\star)$ consists of directions that remain feasible to first order and do not yield a first-order decrease of the objective.

Second-order conditions require a Lagrangian formulation that remains valid even when constraint qualifications fail.

Definition 3.10 (Fritz–John Lagrangian)**.**

$$\mathcal{L}(x, u_0, u, \lambda) = u_0 f(x) + \sum_{i \in \mathcal{P}} \lambda_i h_i(x) + \sum_{j \in \mathcal{Q}} u_j g_j(x),$$

where

$$u_0 \ge 0, \qquad u_j \ge 0 \; (j \in \mathcal{Q}), \qquad \lambda_i \in \mathbb{R}.$$

The coefficient u_0 corresponds to the objective function and may vanish in degenerate (abnormal) cases. Constraint qualifications such as MFCQ exclude this possibility and reduce the

theory to the standard KKT framework.

Theorem 3.11 (Second-Order Necessary Condition for a Local Minimum)**.** *Let $x^\star \in \mathcal{F}$ be a local minimum of problem* (1.1)*. Then for every $d \in D(x^\star)$, there exist multipliers*

$$u_0 \geq 0, \quad u_j \geq 0 \ (j \in \mathcal{Q}), \quad \lambda_i \in \mathbb{R} \ (i \in \mathcal{P}),$$

not all zero, such that

$$\nabla_x \mathcal{L}(x^\star, u_0, u, \lambda) = 0, \tag{3.16}$$

$$d^\top \nabla^2_{xx} \mathcal{L}(x^\star, u_0, u, \lambda)\, d \geq 0, \tag{3.17}$$

$$u_j g_j(x^\star) = 0, \quad j \in \mathcal{Q}, \tag{3.18}$$

$$u_0 \nabla f(x^\star)^\top d = 0, \tag{3.19}$$

$$u_j \nabla g_j(x^\star)^\top d = 0, \quad j \in \mathcal{Q}. \tag{3.20}$$

Proof. Fix an arbitrary direction $d \in D(x^\star)$.

Step 1: First-order optimality and critical directions. Since $x^\star$ is a local minimum of problem (1.1), the Fritz–John necessary conditions apply (Theorem 3.5). Hence, there exist multipliers

$$u_0 \geq 0, \quad u_j \geq 0 \ (j \in \mathcal{Q}), \quad \lambda_i \in \mathbb{R} \ (i \in \mathcal{P}),$$

not all zero, such that the stationarity and complementary slackness conditions (3.16) and (3.18) hold.

By definition of the critical cone $D(x^\star)$, the direction d satisfies

$$\nabla f(x^\star)^\top d \leq 0, \qquad \nabla h_i(x^\star)^\top d = 0 \ (i \in \mathcal{P}), \qquad \nabla g_j(x^\star)^\top d \leq 0 \ (j \in \mathcal{Q}(x^\star)).$$

Thus, d is a feasible first-order direction along which the objective does not increase to first order.

Step 2: Reduction to first-order binding constraints. Define the index set of constraints that are *first-order binding along d*:

$$\mathcal{I}(d) := \{j \in \mathcal{Q}(x^\star) \mid \nabla g_j(x^\star)^\top d = 0\}.$$

Only constraints in $\mathcal{I}(d)$ can affect feasibility at second order. Constraints with $\nabla g_j(x^\star)^\top d < 0$ become strictly inactive for small $t > 0$ and hence do not influence second-order behavior.

The orthogonality conditions

$$u_0 \nabla f(x^\star)^\top d = 0, \qquad u_j \nabla g_j(x^\star)^\top d = 0 \ (j \in \mathcal{Q})$$

express precisely that multipliers may be nonzero only for first-order binding terms.

Step 3: Second-order expansion along feasible curves. Because the constraint functions are twice continuously differentiable, standard feasible-curve arguments guarantee the existence of a twice continuously differentiable curve

$$\gamma : (-\varepsilon, \varepsilon) \to \mathcal{F}$$

such that

$$\gamma(0) = x^\star, \qquad \dot{\gamma}(0) = d.$$

Applying Taylor expansions along $\gamma(t)$ yields

$$\begin{aligned} f(\gamma(t)) &= f(x^\star) + t\, \nabla f(x^\star)^\top d + \frac{t^2}{2}\, d^\top \nabla^2 f(x^\star) d + o(t^2), \\ g_j(\gamma(t)) &= g_j(x^\star) + t\, \nabla g_j(x^\star)^\top d + \frac{t^2}{2}\, d^\top \nabla^2 g_j(x^\star) d + o(t^2). \end{aligned}$$

For all $j \notin \mathcal{I}(d)$, the first-order term $\nabla g_j(x^\star)^\top d < 0$ ensures feasibility for sufficiently small $t > 0$. For $j \in \mathcal{I}(d)$, feasibility depends on the second-order terms.

Step 4: Second-order infeasibility of descent directions. Suppose, for contradiction, that there exist multipliers satisfying (3.16) and (3.18) such that

$$d^\top \nabla^2_{xx} \mathcal{L}(x^\star, u_0, u, \lambda)\, d < 0.$$

Then combining the second-order expansions with complementary slackness yields

$$\mathcal{L}(\gamma(t), u_0, u, \lambda) < \mathcal{L}(x^\star, u_0, u, \lambda) \quad \text{for all sufficiently small } t > 0.$$

Since $u_0 \geq 0$ and all constraints are satisfied along $\gamma(t)$, this strict decrease in the Lagrangian implies

$$f(\gamma(t)) < f(x^\star)$$

for sufficiently small $t > 0$, contradicting that $x^\star$ is a local minimum.

Step 5: Application of a theorem of the alternative. The absence of such a second-order improving feasible direction implies that the system encoding strict descent and feasibility at second order is infeasible. Applying a second-order version of a theorem of the alternative (Farkas or Gordan type) yields the existence of multipliers

$$u_0 \geq 0, \quad u_j \geq 0, \quad \lambda_i \in \mathbb{R},$$

not all zero, satisfying (3.16)–(3.17) together with the stated orthogonality conditions.

Since the direction $d \in D(x^\star)$ was arbitrary, the second-order curvature inequality (3.17) holds for every critical direction. This establishes the second-order necessary condition. □

Theorem 3.12 (Second-Order Necessary Condition under MFCQ)**.** *Let $x^\star$ be a local minimum of problem* (1.1)*. Assume:*

1. *the objective function f and the constraint functions h_i, $i \in \mathcal{P}$, and g_j, $j \in \mathcal{Q}$, are twice continuously differentiable in a neighborhood of $x^\star$;*
2. *the Mangasarian–Fromovitz Constraint Qualification (MFCQ) holds at $x^\star$.*

Then there exist Lagrange multipliers

$$\lambda^\star \in \mathbb{R}^{|\mathcal{P}|}, \qquad \mu^\star \in \mathbb{R}^{|\mathcal{Q}|}_{\geq 0}$$

satisfying the Karush–Kuhn–Tucker conditions at $x^\star$ such that

$$d^\top \nabla^2_{xx} \mathcal{L}(x^\star, \lambda^\star, \mu^\star)\, d \;\geq\; 0 \quad \textit{for all } d \in \mathcal{C}(x^\star), \tag{3.21}$$

where $\mathcal{C}(x^\star)$ denotes the critical cone at $x^\star$.

Proof. Since $x^\star$ is a local minimum and MFCQ holds at $x^\star$, the Karush–Kuhn–Tucker theorem

(Theorem 3.7) guarantees the existence of multipliers $(\lambda^\star, \mu^\star)$ with $\mu^\star \geq 0$ such that

$$\nabla f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i^\star \nabla h_i(x^\star) + \sum_{j \in \mathcal{Q}} \mu_j^\star \nabla g_j(x^\star) = 0,$$

$$\mu_j^\star g_j(x^\star) = 0, \quad j \in \mathcal{Q}.$$

Let $d \in \mathcal{C}(x^\star)$ be arbitrary. By definition of the critical cone, d satisfies

$$\nabla h_i(x^\star)^\top d = 0 \quad (i \in \mathcal{P}),$$

$$\nabla g_j(x^\star)^\top d = 0 \quad (j \in \mathcal{A}^+), \qquad \nabla g_j(x^\star)^\top d \leq 0 \quad (j \in \mathcal{A}^0),$$

where

$$\mathcal{A}^+ := \{j \in \mathcal{Q} \mid \mu_j^\star > 0\}, \qquad \mathcal{A}^0 := \{j \in \mathcal{Q} \mid \mu_j^\star = 0,\ g_j(x^\star) = 0\}.$$

Because f, h_i, and g_j are twice continuously differentiable, there exists a twice continuously differentiable feasible curve $\gamma : (-\varepsilon, \varepsilon) \to \mathcal{F}$ such that

$$\gamma(0) = x^\star, \qquad \dot{\gamma}(0) = d.$$

Such a curve exists because MFCQ ensures a well-defined first-order tangent structure for the feasible set.

Define $\phi(t) := f(\gamma(t))$. Since $x^\star$ is a local minimum, $\phi(t) \geq \phi(0)$ for all sufficiently small t, implying

$$\phi'(0) = 0, \qquad \phi''(0) \geq 0.$$

Using the chain rule and the KKT stationarity condition, we obtain

$$\phi'(0) = \nabla f(x^\star)^\top d = -\sum_{i \in \mathcal{P}} \lambda_i^\star \nabla h_i(x^\star)^\top d - \sum_{j \in \mathcal{Q}} \mu_j^\star \nabla g_j(x^\star)^\top d = 0,$$

where the equality follows from feasibility of d and complementary slackness.

Differentiating a second time and using the feasibility of $\gamma(t)$ yields

$$\phi''(0) = d^\top \left(\nabla^2 f(x^\star) + \sum_{i \in \mathcal{P}} \lambda_i^\star \nabla^2 h_i(x^\star) + \sum_{j \in \mathcal{Q}} \mu_j^\star \nabla^2 g_j(x^\star) \right) d = d^\top \nabla_{xx}^2 \mathcal{L}(x^\star, \lambda^\star, \mu^\star)\, d.$$

Since $\phi''(0) \geq 0$ for every $d \in \mathcal{C}(x^\star)$, the inequality (3.21) follows. This completes the

proof. $\square$

Example 3.13 (Second-order sufficiency under MFCQ for an inequality-constrained problem)**.** Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & f(x) := -x_1 x_2 x_3 \\ \text{s.t.} \quad & g(x) := x_1^2 + 2x_2^2 + 4x_3^2 - 48 \leq 0. \end{aligned} \tag{3.22}$$

Let

$$x^\star = (-4,\, -2\sqrt{2},\, 2).$$

We verify that $x^\star$ is an *isolated local minimum* using the second-order sufficient condition under MFCQ.

Feasibility and activity. A direct computation shows

$$g(x^\star) = (-4)^2 + 2(-2\sqrt{2})^2 + 4(2)^2 = 16 + 16 + 16 = 48,$$

so the inequality constraint is active at $x^\star$. Hence,

$$\mathcal{Q}(x^\star) = \{1\}.$$

Gradients and MFCQ. The gradients of the objective and constraint are

$$\nabla f(x) = \begin{pmatrix} -x_2 x_3 \\ -x_1 x_3 \\ -x_1 x_2 \end{pmatrix}, \qquad \nabla g(x) = \begin{pmatrix} 2x_1 \\ 4x_2 \\ 8x_3 \end{pmatrix}.$$

Evaluating at $x^\star$ gives

$$\nabla f(x^\star) = \begin{pmatrix} 4\sqrt{2} \\ 8 \\ -8\sqrt{2} \end{pmatrix}, \qquad \nabla g(x^\star) = \begin{pmatrix} -8 \\ -8\sqrt{2} \\ 16 \end{pmatrix}.$$

Since $\nabla g(x^\star) \neq 0$, the Mangasarian–Fromovitz Constraint Qualification holds at $x^\star$. Therefore, KKT conditions are necessary for local optimality.

First-order (KKT) stationarity. We seek $\mu \geq 0$ such that

$$\nabla f(x^\star) + \mu \nabla g(x^\star) = 0.$$

Solving componentwise yields

$$\mu = \frac{\sqrt{2}}{2}.$$

Critical directions. The critical cone at $x^\star$ is

$$D(x^\star) = \{d \in \mathbb{R}^3 : \nabla g(x^\star)^\top d = 0\}.$$

That is,

$$-8d_1 - 8\sqrt{2}d_2 + 16d_3 = 0, \quad \text{or equivalently} \quad d_1 + \sqrt{2}d_2 - 2d_3 = 0.$$

Hessian of the Lagrangian. The Hessian of the objective is

$$\nabla^2 f(x) = \begin{pmatrix} 0 & -x_3 & -x_2 \\ -x_3 & 0 & -x_1 \\ -x_2 & -x_1 & 0 \end{pmatrix},$$

and

$$\nabla^2 g(x) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

Thus, the Hessian of the Lagrangian $\mathcal{L}(x, \mu) = f(x) + \mu g(x)$ at $(x^\star, \mu)$ is

$$\nabla^2_{xx} \mathcal{L}(x^\star, \mu) = \begin{pmatrix} 2 & -2\sqrt{2} & 4 \\ -2\sqrt{2} & 4 & 4\sqrt{2} \\ 4 & 4\sqrt{2} & 8 \end{pmatrix}.$$

Restriction to the critical cone. Using the constraint $d_1 = -\sqrt{2}d_2 + 2d_3$, we eliminate d_1 and express the quadratic form in (d_2, d_3):

$$d^\top \nabla^2_{xx} \mathcal{L}(x^\star, \mu)\, d = \begin{pmatrix} d_2 & d_3 \end{pmatrix} \begin{pmatrix} 16 & -8\sqrt{2} \\ -8\sqrt{2} & 32 \end{pmatrix} \begin{pmatrix} d_2 \\ d_3 \end{pmatrix}.$$

The eigenvalues of this matrix are strictly positive. Hence,

$$d^\top \nabla^2_{xx} \mathcal{L}(x^\star, \mu)\, d > 0 \quad \forall d \in D(x^\star) \setminus \{0\}.$$

All assumptions of the second-order sufficient condition under MFCQ are satisfied. Therefore,

$$x^\star = (-4,\ -2\sqrt{2},\ 2)$$

is an *isolated local minimum* of problem (3.22). (In fact, symmetry shows it is one of several global minima.)

4 Further Reading

The study of optimization has long predated its modern mathematical formulation. Philosophical inquiries into "best" arrangements have appeared in ancient Greek thought. For example, Socrates sought general principles explaining how social and natural systems might be organized optimally. A decisive mathematical transition occurred in the seventeenth and eighteenth centuries, when optimization problems emerged naturally from mechanics. Foundational contributions by Galileo, Newton, Leibniz, Euler, Lagrange, and the Bernoulli brothers addressed questions such as maximal projectile range, minimal resistance shapes, and paths of shortest descent. These problems gave rise to the *calculus of variations*, whose central aim is to extremize a functional depending on an entire trajectory rather than on finitely many parameters; see, e.g., [10, 1].

Some constrained optimization problems are even older. Classical *isoperimetric problems*, studied by Zenodorus (ca. 100 BC) and Archimedes, ask which closed curve of fixed length encloses the largest area. These problems already involve implicit constraints and foreshadow modern optimality theory (Encyclopaedia Britannica, 1950). The modern theory of finite-dimensional constrained optimization emerged in the twentieth century, driven by applications in economics, operations research, and engineering. Pioneering work by von Neumann [15] and Kantorovich [7] demonstrated how economic planning and resource allocation problems could be modeled as optimization problems with inequality constraints, particularly linear ones.

First-order optimality conditions for inequality-constrained problems were systematically studied by Karush in his 1939 M.Sc. thesis, written in the context of the Chicago school of calculus of variations; see Kuhn [9]. The most general first-order necessary condition, now known as

the *Fritz–John condition*, was established by Fritz John [6]. Its strengthened form, guaranteeing a nonzero multiplier on the objective function under mild regularity assumptions, was proved by Mangasarian and Fromovitz [11], introducing what is now called the *Mangasarian–Fromovitz Constraint Qualification (MFCQ)*.

By the late 1960s, the classical theory of optimality conditions had largely stabilized. Comprehensive surveys are provided by Mangasarian [11], and systematic treatments of constraint qualifications appear in Peterson [14]. Standard references on nonlinear programming include Avriel [2] and Bazaraa, Sherali, and Shetty [3]. Extensions to nonsmooth optimization require generalized derivatives. In the convex case, the subdifferential provides an adequate replacement for the gradient, while in the nonconvex case Clarke's generalized gradient plays an analogous role; see Clarke [5] and Penot [13]. These developments highlight both the power and the limitations of derivative-based optimality theory.

5 Exercises

Exercise 1 (Local vs. isolated extrema under inequality constraints)**.** Consider the optimization problem

$$\max_{x \in \mathbb{R}^3} f(x) := \frac{x_1}{x_3}$$

subject to

$$\begin{aligned} x_1 + x_2 - 1 &\leq 0, \\ -x_1 - x_2 x_3 + 1 &\leq 0, \\ x_1 \geq 0, \quad x_2 &\geq 0, \quad x_3 \geq \tfrac{1}{2}. \end{aligned}$$

(i) Verify that the point

$$x^\star = (1, 0, 1)$$

is feasible. Identify explicitly which constraints are active at $x^\star$.

(ii) Compute the gradients of the objective function and of all active constraints at $x^\star$. Determine the active set $\mathcal{Q}(x^\star)$.

(iii) Verify that $x^\star$ satisfies the first-order Fritz–John necessary conditions. Explain why these conditions alone do not guarantee local optimality.

(iv) Show that $x^\star$ also satisfies all second-order *necessary* conditions discussed in this chapter, when evaluated along all critical directions.

(v) Consider the feasible curve

$$\gamma(\alpha) = (1, 0, 1 - \alpha)^\top, \qquad \alpha \geq 0.$$

Show that $\gamma(\alpha)$ remains feasible for sufficiently small $\alpha > 0$, and prove that

$$f(\gamma(\alpha)) > f(x^\star).$$

(vi) Conclude that $x^\star$ is *not* a local maximizer, despite satisfying first- and second-order necessary conditions. Briefly explain the geometric reason for this failure.

Exercise 2 (Isolated local maximum via vanishing critical directions)**.** Consider the same optimization problem as in the previous exercise and define

$$y^\star = \left(1, 0, \tfrac{1}{2}\right).$$

(i) Verify that $y^\star$ is feasible and determine the active constraint set $\mathcal{Q}(y^\star)$.

(ii) Compute the set of critical directions $D(y^\star)$ by imposing the first-order feasibility and stationarity conditions.

(iii) Show that

$$D(y^\star) = \{0\}.$$

(iv) Explain why the absence of any nonzero critical direction implies that $y^\star$ is an *isolated local maximum*. Relate your argument to the second-order sufficient condition for constrained optimization.

Exercise 3 (Unconstrained isolated local minimum)**.** Consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^3} f(x),$$

where

$$f(x) = x_1^4 + x_1^3 - x_1 + x_2^4 - x_2^2 + x_2 + x_3^2 - x_3 + x_1 x_2 x_3.$$

(i) Compute $\nabla f(x)$ and verify that the point

$$x^\star = (0.571,\ -0.940,\ 0.768)$$

satisfies $\nabla f(x^\star) = 0$.

(ii) Compute the Hessian matrix $\nabla^2 f(x)$.

(iii) Evaluate $\nabla^2 f(x^\star)$ and show that it is positive definite.

(iv) Conclude that $x^\star$ is an *isolated local minimum* of f. Explain why positive definiteness of the Hessian excludes the existence of any other local minimum in a neighborhood of $x^\star$.

Exercise 4 (Failure of Lagrange multipliers under degeneracy)**.** Consider the equality-constrained problem

$$\min_{x \in \mathbb{R}} f(x) = x \quad \text{s.t.} \quad h(x) = x^2 = 0.$$

(i) Identify the feasible set and show that the unique feasible point $x^\star = 0$ is a global minimizer.

(ii) Compute $\nabla f(x^\star)$ and $\nabla h(x^\star)$.

(iii) Show that the Lagrange multiplier equation

$$\nabla f(x^\star) + \lambda \nabla h(x^\star) = 0$$

has no solution.

(iv) Explain why this does not contradict first-order optimality. Identify the constraint qualification that fails and interpret this failure geometrically.

Exercise 5 (Second-order conditions without constraint qualifications)**.** Consider the inequality-constrained problem

$$\min_{x \in \mathbb{R}} f(x) = x^2 \quad \text{s.t.} \quad g(x) = x^3 \leq 0.$$

(i) Show that $x^\star = 0$ is feasible and is a local minimum.

(ii) Compute $\nabla g(x^\star)$ and verify that the Mangasarian–Fromovitz constraint qualification fails at $x^\star$.

(iii) Determine whether the Karush–Kuhn–Tucker conditions hold at $x^\star$.

(iv) Verify explicitly that the Fritz–John conditions are satisfied.

(v) Decide whether $x^\star$ is an isolated local minimum and explain how this example illustrates the distinction between optimality and regularity.

Exercise 6 (Equality constraints rewritten as inequalities)**.** Consider the equality-constrained problem

$$\min f(x) \quad \text{s.t.} \quad h(x) = 0,$$

and rewrite it equivalently as

$$h(x) \leq 0, \qquad -h(x) \leq 0.$$

(i) Write down the Fritz–John conditions for the reformulated problem.

(ii) Show that all multipliers may vanish at a local optimum.

(iii) Explain why this degeneracy cannot occur when the Mangasarian–Fromovitz constraint qualification holds for the original problem.

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